

Guesstimate

Asked by : Zomato



Problem Statement : Total number of reviews on Zomato ?

Clarifying Questions :

Q1 : What is the duration for which we need to calculate the review (A day/A month/An year or since inception)

Ans : In a year

Q2 : Which reviews do we have to consider app store reviews or in app reviews or something else ?

Ans : In app reviews

Q3 : Do we need to consider reviews from a specific city, state or the whole country ?

Ans : Whole country

Q4 : Do we need to consider reviews from a specific demography of people ?

Ans : No, consider everyone's reviews

Q5 : Reviews through Android or IOS app ?

Ans : Both

Q6 : Since Zomato operates in 15 + countries, do we have to consider all the countries or just India ?

Ans : Consider only India

Approach :

I am following the Top down approach.

Equation : # of reviews on Zomato in a year = Total no. of orders in a year * Avg reviews per order

Now we have to calculate the Total # of orders in a year.

Equation : # of orders in a year = # of people ordering from Zomato * Avg # of orders

Total Population of India = 140 crores

Assumption : Assuming Zomato operates only in urban cities.

Lets divide the population into Urban and Rural.

Assumption : Assuming Urban : Rural split as 30:70

of people living in Urban = 30% * 140 crores = **42 crores**

We have also have to think about how many people have mobile phones to order food.

In Urban areas, lets assume **75% of people have mobile phones with internet connectivity.**

Hence # of people with mobile phones & internet connectivity = 75% * 42 crores = **32 crores** (approx)

Assumption : Lets consider people in the age group of 15-50 years will only order food online.

% of people in this age group = **60%**

Hence, **# of people in the age group (15-50 yrs)** = $60\% * 32 \text{ crores} = 19 \text{ crores}$ (approx.).

Now we have to consider about how many of these people might prefer ordering food online.

Assumption : Assuming 50% of this age group order food online.

Therefore, **# of people ordering food online** = $50\% * 19 \text{ crores} = 10 \text{ crores}$ (approx.)

There are lot of companies in the food delivery business like Swiggy, Dominos etc.

Assumption : Lets consider Zomato has 50% of the market share.

Hence, **# of people ordering through Zomato** = $50\% * 10 \text{ crores} = 5 \text{ crores}$.

Now, we have to calculate orders made through Zomato by these 5 crores.

Categorizing the users into 3 types.

1. Power User - Orders 10 meals per month
2. Active User - Orders 4 meals per month
3. Passive User - Orders 1 meals per month

Assumption : Assuming **30%** of people are the **power users**, **40%** of people are **active users** and another **30%** of the people are **passive users**.

of Power users = $30\% * 5 \text{ crores} = 1.5 \text{ crores}$.

of Active users = $40\% * 5 \text{ crores} = 2 \text{ crores}$.

of Passive users = $30\% * 5 \text{ crores} = 1.5 \text{ crores}$

Total # of orders by Power users = $1.5 \text{ crores} * 10 = 15 \text{ crores orders per month.}$

Total # of orders by Power users in a year = $15 * 12 = 180 \text{ crores orders per year.}$

Total # of orders by Active users = $2 \text{ crores} * 4 = 10 \text{ crores orders per month.}$

Total # of orders by Active users in a year = $10 * 12 = 120 \text{ crores orders per year.}$

Total # of orders by Passive users = $1.5 \text{ crores} * 1 = 1.5 \text{ crores orders per month.}$

Total # of orders by Passive users in a year = $1.5 * 12 = 18 \text{ crores orders per year.}$

Therefore **total orders per year** = $180 + 120 + 18 = 320 \text{ crores order per year}$
approx.

Now we have to calculate **the # of reviews posted.**

We will again categorize into three types.

1. Power reviewer = 10% reviews (10 reviews per 100 orders)
2. Active reviewer = 2% reviews (2 reviews per 100 orders)
3. Passive reviewer = 1% review (1 review per 100 orders)

Average review per order = $(10 + 2 + 1)/(100+100+100) = 0.04 \text{ reviews per order.}$

Total reviews posted on Zomato in a year = Total no. of orders in a year * Avg #
of review per order = $320 \text{ crores} * 0.04 = 12.8 \text{ crores reviews} = 13 \text{ crores approx.}$